

Section A: Conceptual Understanding

1. Explain why REST APIs are preferred for mobile and frontend-heavy apps.

- REST APIs are lightweight and fast.
- They use JSON, which is easy to read and process.
- Frontend and backend can work independently.
- They support multiple platforms like Web, Android, and iOS.

2. Explain how serializers act as a validation layer and how to implement custom field-level validation.

- Serializers convert model data to JSON and JSON to model data.
- They validate data before saving it to the database.
- Custom validation is done using `validate_<fieldname>()`

3. Explain the importance of using appropriate HTTP status codes (201 Created vs 200 OK) for API outcomes.

- Status codes inform clients about request results.
- 200 OK → Request successful.
- 201 Created → New resource successfully created.
- Proper status codes improve API communication and debugging.

4. Explain why pagination is required when listing doctors and how it impacts database performance.

- Prevents loading all records at once.
- Improves API response speed.
- Reduces database load.
- Makes handling large datasets easier.

5. Explain the benefits of ViewSets over APIView for rapid CRUD development.

- Less code to write.
- Built-in CRUD operations.
- Easy router integration.
- Faster API development and maintenance.

6. Explain the use of Atomic Transactions (`transaction.atomic()`) when creating related doctor records to ensure data integrity.

- Ensures database consistency.
- All operations succeed or all fail.
- Prevents partial data saving.
- Maintains data integrity during record creation or updates.